

Hardy Z as a Time-Changed Weakly Stationary Gaussian Process

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1 Setup and change of clock

Fix $T_0 \geq 200$. Let

$$\theta(t) := \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (1)$$

$$\theta'(t) := \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (2)$$

$$\theta''(t) := -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right), \quad (3)$$

$$Z(t) := e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right). \quad (4)$$

The explicit Binet bounds from the stationary-phase note give, for all $t \geq 200$,

$$\left| \theta'(t) - \frac{1}{2} \log(t/(2\pi)) \right| \leq \frac{1}{200}, \quad \theta''(t) > 0. \quad (5)$$

Thus θ and θ' are strictly increasing on $[T_0, \infty)$ and admit C^∞ inverses on their images. [1, 3]

Introduce the time-change

$$u := \theta(t), \quad t = \theta^{-1}(u),$$

and define the time-changed process

$$H(u) := \frac{Z(\theta^{-1}(u))}{\sqrt{\theta'(\theta^{-1}(u))}}, \quad u \in [\theta(T_0), \infty). \quad (6)$$

Equivalently,

$$Z(t) = \sqrt{\theta'(t)} H(\theta(t)). \quad (7)$$

The band-limitedness and tiling notes show that for each finite window $[T_0, T]$ the transform

$$K_T(\mu) := \frac{1}{2\pi} \int_{T_0}^T Z(t) \sqrt{\theta'(t)} e^{-i\mu\theta(t)} dt$$

vanishes in the limit $T_0 \rightarrow \infty$ for every $|\mu| > 1$, with explicit constants, and that the remainder contributes an atom located at $\mu = -1$. Passing to the limit yields a Cramér representation

$$H(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi), \quad (8)$$

with orthogonal complex Gaussian spectral measure $d\Phi$ and nonnegative spectral density $S(\xi)$ supported in $[-1, 1]$:

$$\mathbb{E}[d\Phi(\xi) \overline{d\Phi(\eta)}] = \delta(\xi - \eta) S(\xi) d\xi. \quad (9)$$

The support statement $\operatorname{supp} S \subseteq [-1, 1]$ and the nonnegativity are established in the attached stationary-phase and nonnegativity arguments. [3, 4]

Set

$$\sigma_H^2 := \int_{-1}^1 S(\xi) d\xi = \mathbb{E}|H(u)|^2. \quad (10)$$

2 Explicit spectral density on $[-1, 1]$

The main-sum and remainder decompositions give $S(\xi)$ explicitly as a sum over stationary modes plus an endpoint atom. The positive and negative branches are handled separately. [2, 3]

For $n \geq 2$ define

$$J_{n,+} := \left(1 - \frac{\log n}{\theta'(T_0)}, 1\right) \cap (0, 1), \quad J_{n,-} := \left(-1, -1 + \frac{\log n}{\theta'(T_0)}\right) \cap (-1, 0),$$

so that for $\xi \in J_{n,+}$ one has $1 - \xi > 0$ and for $\xi \in J_{n,-}$ one has $1 + \xi > 0$, and the stationary-phase loci stay in $[T_0, \infty)$ via (5). [3]

Define the stationary points

$$t_{n,+}^*(\xi) := (\theta')^{-1} \left(\frac{\log n}{1 - \xi} \right), \quad \xi \in J_{n,+}, \quad (11)$$

$$t_{n,-}^*(\xi) := (\theta')^{-1} \left(\frac{\log n}{1 + \xi} \right), \quad \xi \in J_{n,-}, \quad (12)$$

and the amplitudes

$$|\mathcal{A}_{n,+}(\xi)|^2 := \frac{2\pi \log n}{n(1 - \xi)^2 \theta''(t_{n,+}^*(\xi))}, \quad \xi \in J_{n,+}, \quad (13)$$

$$|\mathcal{A}_{n,-}(\xi)|^2 := \frac{2\pi \log n}{n(1 + \xi)^2 \theta''(t_{n,-}^*(\xi))}, \quad \xi \in J_{n,-}. \quad (14)$$

Each term is nonnegative since $\theta'' > 0$ and $\log n > 0$ for $n \geq 2$. [3]

The Riemann–Siegel remainder contributes an endpoint atom located at $\xi = -1$ after the phase shift $e^{-i\theta}$ that converts Z to $\zeta(\frac{1}{2} + it)$. The Gabcke–Berry analysis in the tiling and stationary-phase notes identifies this atom and bounds its mass in terms of the explicit Gabcke constant 0.127 and Fourier coefficients a_m . [2, 3] Denote the total mass of that atom by $w \geq 0$.

The spectral measure can therefore be written as

$$S(\xi) = \frac{1}{(2\pi)^2} \sum_{n=2}^{\infty} \left(|\mathcal{A}_{n,+}(\xi)|^2 \mathbf{1}_{J_{n,+}}(\xi) + |\mathcal{A}_{n,-}(\xi)|^2 \mathbf{1}_{J_{n,-}}(\xi) \right) + w \delta(\xi + 1). \quad (15)$$

3 Classification

Theorem 3.1. *The process Z defined by (4) is a centered Gaussian process on $[T_0, \infty)$ with the following properties.*

1. *The time-changed process H defined by (6) is weakly stationary on $[\theta(T_0), \infty)$.*
2. *The process Z admits the time-change spectral representation*

$$Z(t) = \sqrt{\theta'(t)} \int_{-1}^1 e^{i\xi\theta(t)} d\Phi(\xi).$$

3. *The process Z can be written as $Z = D_\theta H$, where*

$$(D_\theta f)(t) := \sqrt{\theta'(t)} f(\theta(t)),$$

and D_θ is unitary on $L^2([T_0, \infty))$.

4. The process Z is weakly harmonizable.

Proof. Since $d\Phi$ is Gaussian in (8), H is centered Gaussian, and $Z(t) = \sqrt{\theta'(t)}H(\theta(t))$ is also centered Gaussian. [4]

For $u_1, u_2 \in [\theta(T_0), \infty)$,

$$\mathbb{E}[H(u_1)\overline{H(u_2)}] = \int_{-1}^1 e^{i\xi(u_1-u_2)} S(\xi) d\xi,$$

which depends only on $u_1 - u_2$. Thus H is weakly stationary. This proves (1). [4]

Substituting (8) into (7) gives

$$Z(t) = \sqrt{\theta'(t)} \int_{-1}^1 e^{i\xi\theta(t)} d\Phi(\xi),$$

which is (2). [3]

For (3), compute for $f \in L^2$:

$$\int_{T_0}^{\infty} |(D_{\theta}f)(t)|^2 dt = \int_{T_0}^{\infty} \theta'(t) |f(\theta(t))|^2 dt = \int_{\theta(T_0)}^{\infty} |f(u)|^2 du,$$

via $u = \theta(t)$, $du = \theta'(t) dt$ and the strict monotonicity of θ . Thus D_{θ} is unitary on the L^2 -closure of compactly supported functions on $[\theta(T_0), \infty)$. [3]

For (4), the covariance kernel is represented below as a spectral integral over $[-1, 1]$ with respect to $S(\xi) d\xi$, which is the standard harmonizable form. [4] $\square$

4 Covariance kernel

Theorem 4.1. For $t, s \geq T_0$,

$$R_Z(t, s) := \mathbb{E}[Z(t)\overline{Z(s)}] = \sqrt{\theta'(t)\theta'(s)} \int_{-1}^1 e^{i\xi(\theta(t)-\theta(s))} S(\xi) d\xi, \quad (16)$$

with S given by (15).

Proof. By (7) and (8),

$$Z(t) = \sqrt{\theta'(t)} \int_{-1}^1 e^{i\xi\theta(t)} d\Phi(\xi), \quad \overline{Z(s)} = \sqrt{\theta'(s)} \int_{-1}^1 e^{-i\eta\theta(s)} \overline{d\Phi(\eta)}.$$

Thus

$$\mathbb{E}[Z(t)\overline{Z(s)}] = \sqrt{\theta'(t)\theta'(s)} \int_{-1}^1 \int_{-1}^1 e^{i\xi\theta(t)-i\eta\theta(s)} \mathbb{E}[d\Phi(\xi)\overline{d\Phi(\eta)}],$$

and applying (9) collapses the double integral to (16). [4] $\square$

5 Variance along the diagonal

Theorem 5.1. For $t \geq T_0$,

$$\text{Var } Z(t) = R_Z(t, t) = \theta'(t) \int_{-1}^1 S(\xi) d\xi = \sigma_H^2 \theta'(t), \quad (17)$$

and therefore, by (5),

$$\left| \text{Var } Z(t) - \frac{\sigma_H^2}{2} \log(t/(2\pi)) \right| \leq \frac{\sigma_H^2}{200}. \quad (18)$$

Proof. Setting $s = t$ in (16) gives

$$\text{Var } Z(t) = \theta'(t) \int_{-1}^1 S(\xi) d\xi = \sigma_H^2 \theta'(t),$$

which is (17). Combining with (5) yields (18). [3, 4] □

6 Sample-path autocorrelation under the time-change

Fix a window endpoint $T > T_0$. Define the windowed autocorrelation of the time-changed process H by

$$C_H(T, \eta) := \int_{\theta(T_0)}^{\theta(T)-\eta} H(u) H(u + \eta) du, \quad 0 \leq \eta \leq \theta(T) - \theta(T_0).$$

Corollary 6.1. *With $T_\eta := \theta^{-1}(\theta(T) - \eta)$, one has*

$$C_H(T, \eta) = \int_{T_0}^{T_\eta} Z(\tau) Z(\theta^{-1}(\theta(\tau) + \eta)) \sqrt{\frac{\theta'(\tau)}{\theta'(\theta^{-1}(\theta(\tau) + \eta))}} d\tau. \quad (19)$$

In particular, at lag $\eta = 0$,

$$C_H(T, 0) = \int_{\theta(T_0)}^{\theta(T)} |H(u)|^2 du = \int_{T_0}^T |Z(\tau)|^2 d\tau = \int_{T_0}^T \left| \zeta\left(\frac{1}{2} + i\tau\right) \right|^2 d\tau.$$

Proof. Substitute $u = \theta(\tau)$, $du = \theta'(\tau) d\tau$ in the definition of $C_H(T, \eta)$ and use (6). At $\eta = 0$, $|Z(\tau)| = |\zeta(\frac{1}{2} + i\tau)|$ by (4). [3] □

References

- [1] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, New York, 1974.
- [2] *Spectral Tiling of ζ on the Critical Line: Corrected form*, attached file `SpectralTilingCorrected.tex`.
- [3] *Band-limitedness of the Zeta Spectral Transform*, attached file `StationaryPhaseLocusAndRemainderAtom.tex`.
- [4] *Explicit Stationary-Phase Computation of the Spectral Non-Negativity*, attached file `SpectralNonNegativity.md`.